

Chidera Onyebu

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Backend and systems engineer: a deployed payments marketplace, an LLM agent whose safety gate is enforced by the graph rather than the prompt, and compilers and garbage collectors in C, OCaml, and Rust.

EDUCATION

University of Maryland, College Park

College Park, MD

B.S. Computer Science and B.S. Applied Mathematics — GPA 3.7/4.0

Aug. 2024 – May 2028

- **Honors: Three-time Gold Medalist, Nigerian National Mathematics Olympiad**; 1st Place (Senior Category), Rotary Mathematics Competition, 2024; Alpha Lambda Delta.
- **Coursework:** Algorithms, Data Structures, Computer Systems (C, x86-64), Organization of Programming Languages (OCaml, Rust), Probability & Statistics, Linear Algebra.

PROJECTS

ToolBelt — Two-Sided Gig Marketplace | *FastAPI, PostgreSQL, TypeScript, Stripe, Docker* Jul. 2026 – Present

- Built and deployed a two-sided marketplace live on a public domain over TLS: an 8,400-line FastAPI/PostgreSQL REST API and a 4,400-line TypeScript Expo client shipping from one codebase to iOS, Android, and web.
- Eliminated a double-charge defect in which Stripe's 24-hour idempotency-key expiry turned a retried refund into a second real charge, by journaling every money-moving call on a separate connection *before* the call is issued.
- Implemented and validated a double-entry ledger across authorize, capture, and payout against live Stripe test mode at a zero trial balance, with a reconciliation sweeper that is read-only by construction.
- Fixed three defects that surface only against the live API — remote capture with local rollback, payout failure undoing job completion, idempotency-key poisoning — each now regression-tested; **228 API and 392 client tests** in CI.

TradeDesk — LLM Agent Platform | *Python, LangGraph, FastAPI, MCP, pytest*

Jun. – Aug. 2026

- Made an unconfirmed trade **structurally impossible**: the compiled LangGraph state machine contains no trade-to-confirmation edge and a test asserts its absence, so no prompt or handler change can bypass it.
- Validated the LangGraph rewrite with a differential-test oracle: retained the hand-rolled orchestrator alongside it and required 12 scripted conversations to agree across both on replies, citations, pending state, and broker side effects.
- Proved the gate under attack: 18 adversarial evaluations (prompt injection, privilege escalation, scope) asserted at the broker through a spy pass **18/18 with 0 unauthorized executions**, across **293 tests**.

Compilers, Runtimes, and Systems Programming | *C, OCaml, Rust*

2025 – 2026

- Implemented the recursive executor for a UNIX shell in C, walking the parse tree to run pipelines, **&&** chains, and subshells over **fork/exec** with **dup2** redirection, exit-status propagation, and ambiguous-redirect detection.
- Built a front-to-back toolchain for a C subset in OCaml (1,700+ lines) — lexer, parser, interpreter, static type checker, stack-machine compiler — plus regex-to-NFA-to-DFA construction validated by a 100-case property test.
- Wrote three garbage collectors in Rust over one shared heap and reachability pass: reference-counting, mark-and-sweep, and semispace stop-and-copy that relocates live objects and rewrites both child pointers and stack roots.

EXPERIENCE

Capstone Engineer, AI.Accelerate FY26

Jun. – Aug. 2026

ServiceNow (external program)

Remote

- Built and demoed TradeDesk, a self-proposed agentic trading assistant, live at Discover ServiceNow as the capstone of a 12-session program, backed by an 8-risk safety audit in which every claimed control cites a named passing test.

Machine Learning Contributor, AI4ALL

Jun. – Aug. 2026

Apply AI Program — 4-person team

Remote

- Shipped the deployed inference app for a 4-person image-model team and merged it upstream, then caught a capture-session leak their own check missed: **94.7% of test examples** shared a session with training data, inflating accuracy by **3.3 points**.

Peer Mentor and Academic Mentor

Sept. 2024 – Present

ColorStack & Alpha Lambda Delta Honor Society, UMD

College Park, MD

- Coach 50+ students one-on-one and lead 18+ debugging workshops and review sessions, teaching problem decomposition rather than supplying answers.

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C, OCaml, Rust, SQL, HTML/CSS

Frameworks: FastAPI, REST APIs, LangGraph, React, React Native/Expo, Node.js, Pydantic, SQLAlchemy, pytest, Jest

Infrastructure: PostgreSQL, Redis, Docker, Stripe, Render, Cloudflare, GitHub Actions CI/CD, Git, UNIX

Practices: mypy --strict, unit, integration, and property-based testing, schema migrations, idempotency design